

Ex.no:9	MINI PROJECT – IPL SPORTS ANALYTICS DATA ANALYTICS

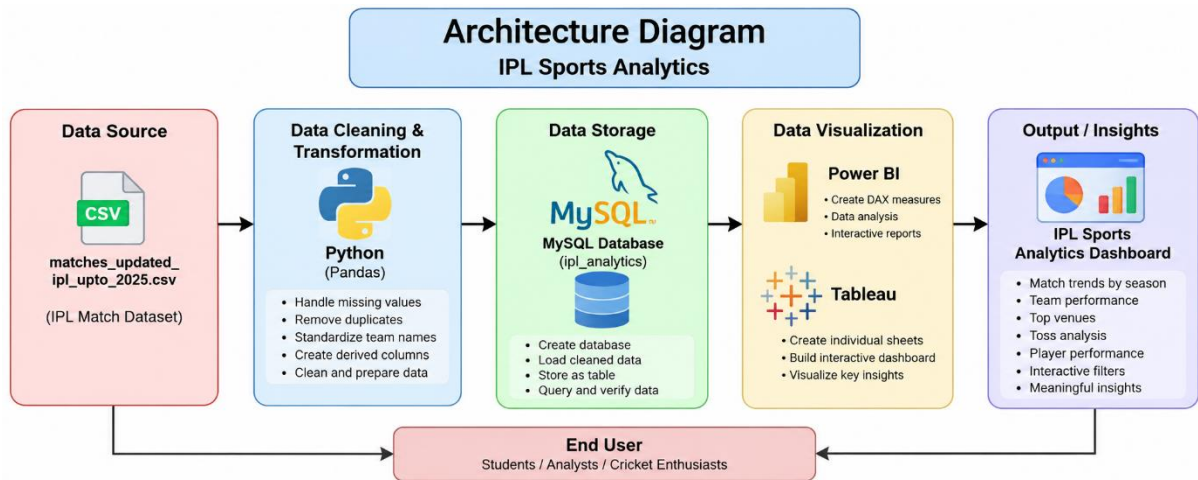
Aim:

To clean, transform, store, and analyze IPL cricket match data using Python, MySQL, Power BI and Tableau to derive meaningful insights into match trends, team performance, venue statistics, player performance, toss decisions, and match outcomes.

Algorithm:

1. Load the IPL match dataset from the CSV file using Pandas.
2. Clean the data by handling missing values, removing duplicate records, and eliminating unnecessary attributes.
3. Transform the data by standardizing team names and creating useful derived fields such as Season Sort and Toss Match Result.
4. Store the cleaned and transformed dataset in a MySQL database named ipl_analytics.
5. Connect the MySQL database to Power BI for data analysis and reporting.
6. Create DAX measures for key metrics such as total matches, total seasons, total teams, and toss winner success percentage.
7. Develop interactive visualizations to analyze IPL matches by season, matches won by teams, top IPL venues, toss decisions, toss winner results, and player of the match awards.
8. Connect the cleaned IPL dataset to Tableau and create separate worksheets for match trends, team performance, venue analysis, toss analysis, and player performance.
9. Generate meaningful sports analytics insights from the Power BI and Tableau dashboards.
10. End the process.

Architecture:



Program:

1. Python – Data Cleaning & Transformation

```
import pandas as pd
```

```
from sqlalchemy import create_engine, text
```

```
# Load dataset
```

```
df = pd.read_csv("matches_updated_ipl_upto_2025.csv")
```

```
# Create a copy for processing
```

```
ipl = df.copy()
```

```
# Display original shape
```

```
print("Original Shape:", ipl.shape)
```

```
# Remove unnecessary columns if available
```

```
unwanted_columns = [
```

```
    "umpire1",
```

```
    "umpire2",
```

```
    "umpire3",
```

```

        "method",
        "result"
    ]

ipl.drop(
    columns=[col for col in unwanted_columns if col in ipl.columns],
    inplace=True
)

# Handle missing values
ipl["city"] = ipl["city"].fillna("Unknown")
ipl["venue"] = ipl["venue"].fillna("Unknown")
ipl["winner"] = ipl["winner"].fillna("No Result")

# Remove duplicate records
ipl.drop_duplicates(inplace=True)

# Standardize team names
team_mapping = {
    "Kings XI Punjab": "Punjab Kings",
    "Delhi Daredevils": "Delhi Capitals",
    "Royal Challengers Bangalore":
        "Royal Challengers Bengaluru"
}

for column in ["team1", "team2", "toss_winner", "winner"]:
    if column in ipl.columns:
        ipl[column] = ipl[column].replace(team_mapping)

# Convert date column
ipl["date"] = pd.to_datetime(
    ipl["date"],

```

```
        errors="coerce"
    )

# Create Season Sort column
def season_sort(season):
    season = str(season)

    if "/" in season:
        return float(season[:4]) + 0.5
    else:
        return float(season)

ipl["Season Sort"] = ipl["season"].apply(
    season_sort
)

# Create Toss Match Result
def toss_match_result(row):

    if pd.isna(row["winner"]) or row["winner"] == "No Result":
        return "No Result"

    if row["toss_winner"] == row["winner"]:
        return "Toss Winner Won"

    return "Toss Winner Lost"

ipl["Toss Match Result"] = ipl.apply(
    toss_match_result,
    axis=1
)
```

```
# Standardize column names
```

```
ipl.columns = (  
    ipl.columns  
    .str.strip()  
    .str.lower()  
    .str.replace(" ", "_")  
)
```

```
# Display cleaned information
```

```
print("Cleaned Shape:", ipl.shape)
```

```
print("\nMissing Values:")
```

```
print(ipl.isnull().sum())
```

```
print("\nFirst 5 Records:")
```

```
print(ipl.head())
```

Original Shape: (1195, 21)
Cleaned Shape: (1195, 17)

Missing Values:

	0
id	0
season	0
city	0
date	0
team1	0
team2	0
dtype:	int64

First 5 Records:

	id	season	city	date	team1	team2	toss_winner	toss_decision	winner	venue	Season Sort	Toss Match Result
0	1	2008	Bangalore	2008-04-18	Royal Challengers Bangalore	Kolkata Knight Riders	Kolkata Knight Riders	bat	Kolkata Knight Riders	M Chinnaswamy Stadium	2008.0	Toss Winner Won
1	2	2008	Chandigarh	2008-04-19	Kings XI Punjab	Chennai Super Kings	Chennai Super Kings	bat	Chennai Super Kings	Punjab Cricket Association Stadium	2008.0	Toss Winner Won
2	3	2008	Delhi	2008-04-19	Delhi Daredevils	Rajasthan Royals	Rajasthan Royals	bat	Rajasthan Royals	Feroz Shah Kotla	2008.0	Toss Winner Won
3	4	2008	Mumbai	2008-04-20	Mumbai Indians	Royal Challengers Bangalore	Mumbai Indians	bat	Mumbai Indians	Wankhede Stadium	2008.0	Toss Winner Won
4	5	2008	Kolkata	2008-04-20	Kolkata Knight Riders	Deccan Chargers	Kolkata Knight Riders	bat	Kolkata Knight Riders	Eden Gardens	2008.0	Toss Winner Won

2. MySQL – Database and Data Loading

MySQL Command Prompt:

```
net start MySQL80
```

```
mysql -u root -p
```

MySQL:

```
CREATE DATABASE IF NOT EXISTS ipl_analytics;
```

Python/Jupyter:

```
username = "root"
```

```
password = "YOUR_MYSQL_PASSWORD"
```

```
host = "localhost"
```

```
port = 3306
```

```
# Connect to MySQL database
```

```
db_engine = create_engine(  
    f'mysql+pymysql://{username}:{password}@'  
    f'{host}:{port}/ipl_analytics'  
)
```

```
# Load cleaned data into MySQL
```

```
ipl.to_sql(  
    "ipl_matches",  
    con=db_engine,  
    if_exists="replace",  
    index=False,  
    chunksize=5000  
)
```

```
print("Data loaded successfully into MySQL.")
```

Verify:

```
pd.read_sql(  
    "SELECT COUNT(*) AS total_matches "  
    "FROM ipl_matches",  
    db_engine  
)
```

```
MySQL 8.0 Command Line Client

C:\>net start MySQL80
The MySQL80 service was started successfully.

C:\>mysql -u root -p
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.0.36 MySQL Community Server - GPL

Copyright (c) 2000, 2024, Oracle and/or its affiliates.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE IF NOT EXISTS ipl_analytics;
Query OK, 1 row affected (0.01 sec)

mysql> USE ipl_analytics;
Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_ipl_analytics |
+-----+
| ipl_matches              |
+-----+
1 row in set (0.00 sec)

mysql> SELECT COUNT(*) AS total_matches FROM ipl_matches;
+-----+
| total_matches |
+-----+
|          1195 |
+-----+
1 row in set (0.00 sec)

mysql> 
```

3. Power BI – DAX Measures

Total Matches =COUNTROWS('ipl_matches')

Total Seasons =DISTINCTCOUNT('ipl_matches'[season])

Total Teams =

```
COUNTROWS(
    DISTINCT(
        UNION(
            SELECTCOLUMNS(
                'ipl_matches',
                "Team",
                'ipl_matches'[team1]
            ),
            SELECTCOLUMNS(
                'ipl_matches',
                "Team",
                'ipl_matches'[team2]
            )
        )
    )
)
```

```

    )
    )
Toss Winner Wins =
CALCULATE(
    COUNTROWS('ipl_matches'),
    'ipl_matches'[Toss Match Result] =
        "Toss Winner Won"
)

```

The screenshot shows the Power BI Desktop interface. The 'Table tools' ribbon is active, showing the 'Name' property set to 'Toss Winner Wins', the 'Home table' set to 'matches_updated_i...', and the 'Format' set to 'Whole number'. The 'Structure' pane on the left shows a visual for 'Total Matches' with a value of 1K. The 'DAX' pane on the right shows the following formula:

```

1 Toss Winner Wins =
2 CALCULATE(
3     COUNTROWS('matches_updated_ipl_upto_2025'),
4     FILTER(
5         'matches_updated_ipl_upto_2025',
6         'matches_updated_ipl_upto_2025'[Toss Winner]
7         =
8         'matches_updated_ipl_upto_2025'[winner]
9     )
10 )

```

```

Toss Winner Success % =
DIVIDE(
    [Toss Winner Wins],
    [Total Matches],
    0
)

```

The screenshot shows the Power BI Desktop interface. The 'Table tools' ribbon is active, showing the 'Name' property set to 'Toss Winner Success...', the 'Home table' set to 'matches_updated_i...', and the 'Format' set to 'Percentage'. The 'Structure' pane on the left shows a visual for 'Total Matches' with a value of 1K. The 'DAX' pane on the right shows the following formula:

```

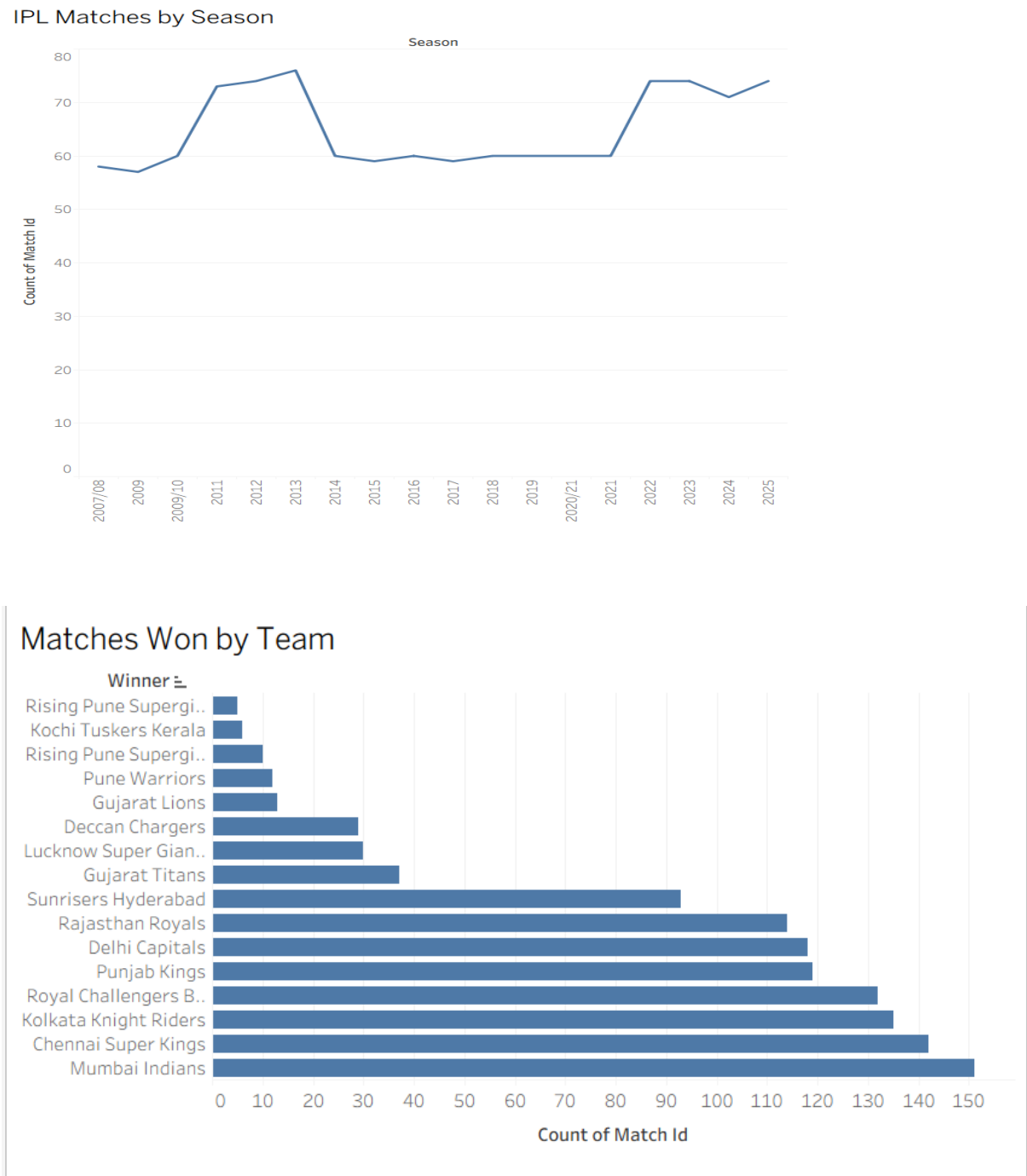
1 Toss Winner Success % =
2 DIVIDE(
3     [Toss Winner Wins],
4     COUNTROWS(
5         FILTER(
6             'matches_updated_ipl_upto_2025',
7             NOT ISBLANK('matches_updated_ipl_upto_2025'[winner])
8         )
9     ),
10     0
11 )

```

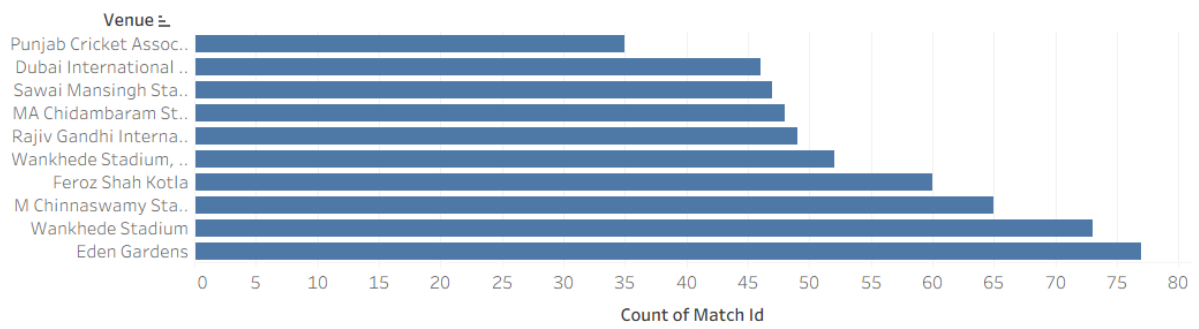
4. Tableau Dashboard

Created visuals in Tableau in separate sheets and created a dashboard for IPL sports performance analysis.

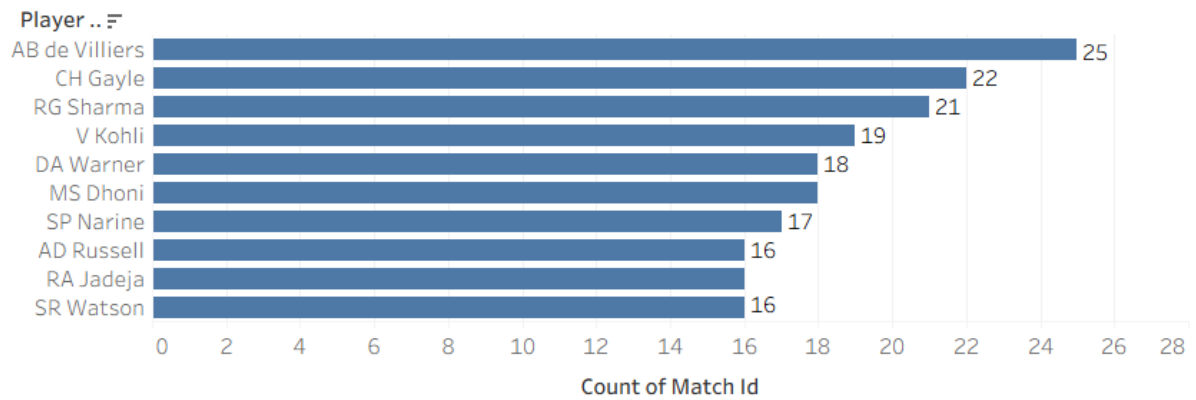
Output:

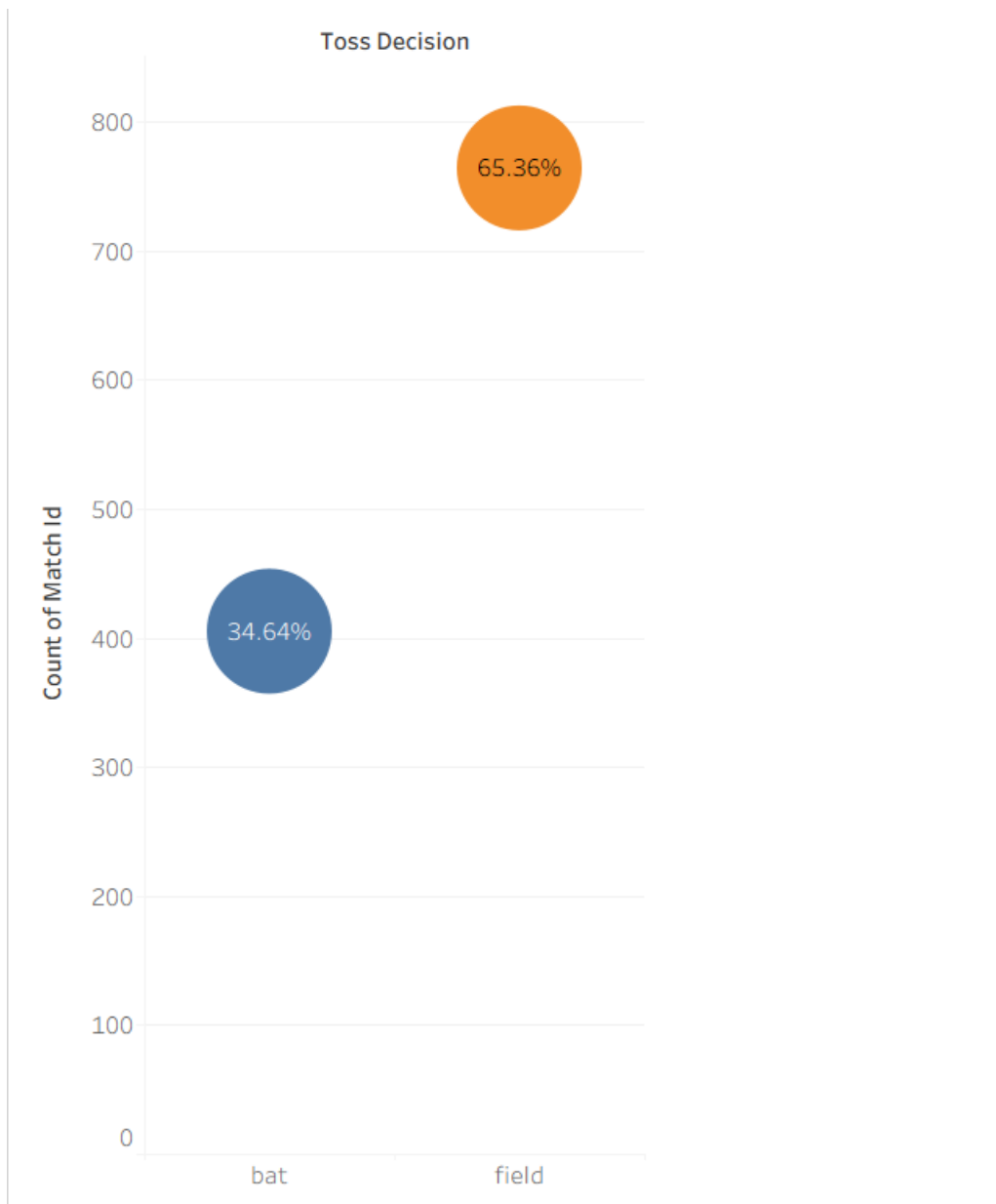


Top 10 IPL Venues by Matches

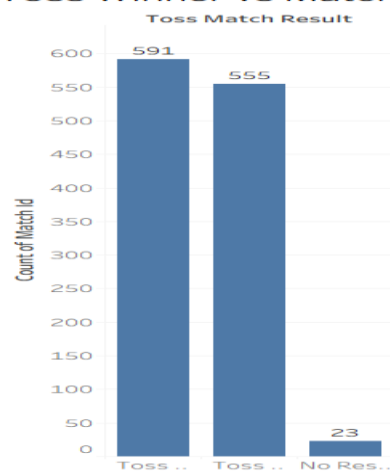


Sheet 6



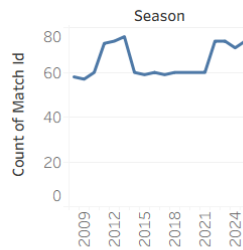


Toss Winner vs Match Winner

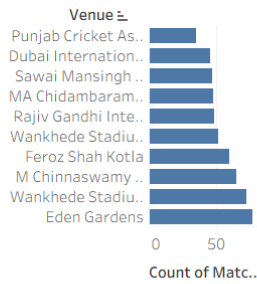


IPL SPORTS PERFORMANCE ANALYTICS

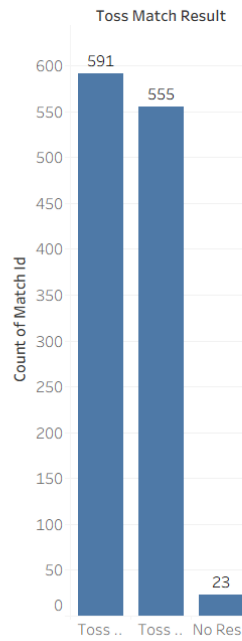
IPL Matches by Season



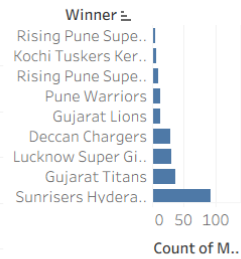
Top 10 IPL Venues by Matches



Toss Winner vs Match Winner



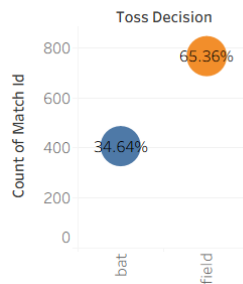
Matches Won by Team



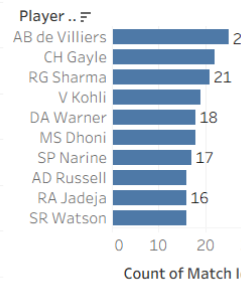
Winner

- ☐ (All)
- ☒ Not
- ☐ Chennai Sup...
- ☐ Deccan Char...
- ☐ Delhi Capitals
- ☐ Gujarat Lions
- ☐ Gujarat Titans
- ☐ Kochi Tusker...
- ☐ Kolkata Knig...
- ☐ Lucknow Su...
- ☐ Mumbai Indi...
- ☐ Pune Warriors
- ☐ Punjab Kings

Toss Decision Distribution



Sheet 6



Toss Decision

- ☐ bat
- ☒ field

Player Of Match

- ☐ (All)
- ☒ Not
- ☐ A Chandila
- ☐ A Kumble
- ☐ A Manohar
- ☐ A Mishra
- ☐ A Nehra
- ☐ A Nortje
- ☐ A Singh
- ☐ A Symonds

Limit

Top 10 by
COUNT([Match Id])

Result: